

Analytical Reflectance Spectroscopic Method for Analysis of Iron Oxide Lake Dye Coating Solutions

Zubair Uddin Siddiqui^{1,*} and S. Arif Kazmi²

¹Department of Chemistry, University of Karachi, Pakistan and ²Department of Bio-inorganic, HEJ Research Institute of Chemistry, University of Karachi, Karachi, Pakistan

Abstract: A dye is a colored compound normally used in solution which is capable of being fixed to a fabric, leather, pharmaceutical granular and tablets etc. A dye is a distinct chemical material which exhibits coloring power when dissolved. A lake is insoluble in nearly all solvents generally is an insoluble material and are more stable than the corresponding water soluble dyes. The lake consists of aluminum hydroxide as substrate.

For quantitative analysis by absorption spectroscopy dyes should be clear and homogenous. In case of lake dyes, absorption spectroscopy is not suitable as a tool for quantitative measurement. In this study we used reflectance spectroscopy and measured lake dyes in pharmaceutical tablets coating solution. In the current study a new methods of analyzing lake dyes is describes using reflectance spectroscopy.

In this project pharmaceutical lake iron oxide coating solution is measured very precisely and analytical works performed using ultrascan xenon equipment with xenon detector under the wave length range from 360 nm to 750 nm with wave length accuracy of 0.75 nm.

Iron oxide red lake coating solution is analyzed and use specially design stainless steel 316 grade trays having area 37.8 cm² and pour lake coating solution on the surface of stainless steel tray and dry it at ambient temperature for two hours and measure its reflectance against the standard prepared in the same manner, using fifty different batches of iron oxide red lake coating solutions and determined random error which is under two percent tolerance limit.

The proposed method is rapid, precise and the obtained results are in a good agreement with the declared content. The precision of the method has been confirmed by the statistical parameter

In conclusion, the present analytical study shows reliability of data and high degree of precision under our define new analytical method for the measurement of iron oxide red lake dye coating solution in terms of reflectance.

Key Words: Lake iron oxide red, 316 grade stainless steel trays, Ultrascanxenon determination.

INTRODUCTION

The fundamental sample requirement for quantitative analysis by spectrophotometry using Beer's law relationship between absorbance and concentration is a clear homogeneous solution. This requirement is not met by lake dyes.

In general lakes dyes are more stable than the corresponding water soluble dyes producing more stable color and are most suitable for product containing oil and fats like vitamin E and vitamin D oil containing products in pharmaceuticals. In general lakes produced in the form of fine powder, lakes are fast to light, air and moisture resist as well as insoluble in alcohol, water and organic solvents. In pharmaceutical industry color is used as indicator of quality, it is also used to differentiate between medicines, to indicate test result on dip strip, part of a brand identity, ease of swallow, dye is used to minimize bitter taste, stomach safety through enteric coating and an indicator of degradation of loss of potency over time and in quality control batch to batch color consistency is an important indicator of quality

before they are shipped to medical stores for sale to end users.

Traditional methods of analysis of lake dyes using optical and spectrophotometric methods for transmittance measurement suffer from the disadvantages that of settling down of dyes particles before proper operation of instrument and made it inappropriate and inaccurate analyzing methods.

Aluminum lakes are produced by the absorption of water soluble dyes on to a hydrated aluminum substrate rendering the color insoluble in water. The end product is colored either by the dispersion of the lake in to the product or by coating on to the surface of the product. A dye is a distinct chemical material, which exhibits coloring power when dissolved, generally lake dyes are insoluble material which color by its dispersion consist of a substrate of alumina hydrate on which the dye is absorbed or precipitated.

Visible region 400 nm to 750 nm consists of electromagnetic radiation and every wavelength consists of definite energy level and prepares particular color when it strikes the eye. Whenever visible light consist of complete wavelength its appear white and absence of visible region produce darkness or black.

*Address correspondence to this author at the Department of Chemistry, University of Karachi, Pakistan; E-mail: zubairprac@gmail.com

When visible light hit a color material some wavelength are absorbed and other are reflected and actually reflected wavelength compose the color of the material. If a material absorbs wavelength in green-blue region its appeared red which is the color of reflected wavelength.

Table 1. Correlates the Wavelength Absorbed and Reflected Color Observed

Color Observed	Wavelength Absorbed	Reflected Color
Violet	400nm to 435nm	Yellow – Green
Blue	435nm to 480nm	Yellow
Blue – Green	480nm to 490nm	Orange
Green – Blue	490nm to 500nm	Red
Green	500nm to 560nm	Purple
Yellow – Green	560nm to 580nm	Violet
Yellow	580nm to 595nm	Blue
Orange	595nm to 605nm	Blue – Green
Red	605nm to 750nm	Green – Blue

In conclusion, the present analytical study shows reliability of data and high degree of precision under our define new analytical method for the measurement of iron oxide red lake dye coating solution in terms of reflectance.

REFLECTANCE SPECTROSCOPY

When an object is irradiated with electromagnetic radiation the object absorb whatever wavelengths it is capable of absorbing, the unabsorbed light is reflected and quantities by optical detector. When our eyes perceive a red substance we are actually observing the reflected portion of the irradiation light. The red substance absorbed blue light, reflect the unabsorbed red light, our eyes perceive the reflected red light, and our mind then interprets this measurement to conclude that the actually red color.

Most analytical chemists are aware with ultraviolet and visible spectrophotometers for the measurement of absorbance or transmittance as well as wave length scanning in UV/visible absorbing analytes in solution. The basic principles of UV/Visible spectroscopy is well known as light is generated by a lamp which is used normally tungsten lamp for visible region of spectrum and deuterium lamp for the ultraviolet region. The light is dispersed in to its constituent wavelengths in a monochromator which results in a narrow band of the dispersed spectrum passing from exit slit of monochromator. Suitable optics are used to lead this light, of a narrow bandwidth, to the sample to be measured. A UV/visible absorbing sample absorb certain amount of light and remaining light is detected by a suitable detector.

The Beer-Lambart law is then applied to determine concentration of specific analyte in the sample at specific wavelength.

Beer-Lambart law describe linear relationship between absorbance and concentration but some restriction of law and the linearity of Beer-Lambart law is limited by chemical and instrumental factors. Causes of non linearity include:

- Deviation in molar absorptivity coefficient at high concentration ($> 0.01M$) due to electrostatic interaction between molecules in close proximity.
- Scattering of light due to particulate in the sample.
- Fluorescence or phosphorescence of the samples.
- Stray light effect (other than selected nanometer light reaching the detector).
- Non-monochromatic radiation.
- Changes in refractive index at high analyte concentration.
- Shift in chemical equilibria as function of concentration.
- Very large and complex molecules and non homogeneous and non clear sample.

In the current analytical study method of lake dye coating solution has been described using reflectance spectroscopy. Reflectance measurement is of great value in providing a reference standard for the comparison of the color of different samples.

A reflectance spectrophotometer is similar to UV/visible spectrophotometer it consist narrow band width to provide well resolved visible spectra yet wide enough to provide a good energy level for diffuse reflectance measurement. The adopted reflectance spectrophotometer ultrascan xenon spectrophotometer able to make measurement both at selected fixed wavelength or perform scans over the complete wavelength range. In color chemistry, color cannot describe solely in spectroscopic terms three attribute commonly used to identify a color:

Value

This attribute of color distinguish a light color from a dark one. Value is the lightness or darkness of a color, as it has to do with the addition of white or black and not with the mixing of color, sometime called tone.

Chroma

The quality that distinguish a strong color from a weak color. Chroma is a Greek word for color, may refer to colorfulness or chroma, the perceived intensity of a specific color.

Hue

The quality by which one color family is distinguish from another color family. Hue is one of the main property of a color, defined technically, as the degree to which a stimulus

can be described as similar to or different from stimuli that are described as red, green, blue, and yellow.

The three attribute of color may be used to define a three dimension color in which any color is located by its coordinates. The observed color of an object depend on the spectral energy of the illumination, the observing characteristic of the object and the visual sensitivity of the observer over the visible range.



Fig. (1). Stainless steel 316 trays for coating solution measurement.

Instrumental method for measurement of color provides more objective data than the subjective viewing of color by a small number of individuals.

MATERIAL AND METHODS

Analytical experiment were performed using an Ultra scan XE colorimeter system with detector Xenon flash spectrophotometer with a visible range with wave length accuracy of 0.75 nm. Colorimeter work and based spectral colors red, green, yellow, blue, darkness & brightness in respectively X, Y & Z axis. To check the response of the instrument, standard accessories (white calibrated tile, green tile, light trap, black card, black card device, didymium filter) were used through reflectance port.

Instrumental name : Ultrascan Colorimeter XE.
 Model No. : USXE/SAV/UV- 2
 Manufacturer's Name : Hunter Association Laboratory, Inc.
 Manufacturer's Address : 11491 Sunset Hills Road, Reston, Virginia U.S.A. 20190



Fig. (2). Ultrascan Xenon Instrument.

PREPARATION OF STANDARD AND SAMPLE

Sample Preparation

We take 2.986 ml coating solution of dye Red lake code No. 25880 on to the SST trays, spread the solution and dry about two hours at ambient temperature. This is assay sample preparation.

Standard Preparation

We take 2.986 ml coating solution of dye Red lake code No. 25880 from any previous approved lot on to the SST tray, spread the solution and dry about two hours at ambient temperature. This is assay standard preparation.

We Place the assay standard and assay sample SST tray on the port plate of Ultrascan colorimeter XE and measured the reading in reflectance mode & calculate ΔE .

$$\Delta E = \sqrt{(L)^2 + (A)^2 + (B)^2}$$

L = The difference between sample and standard values in X-axis.

A = The difference between sample and standard values in Y-axis

B = The difference between sample and standard values in Z-axis

ΔE = Total difference in X, Y and Z axis against standard values.

VALIDATION

Specificity

Dye Red Code No. 25880

Control No. 80648XW

Dye Red Code No. 26115

Control No. 07574XW

38 ml of 0.02 % dye red coating solution code 25880 is mixed with 2 ml of 0.02 % dye red coating solution code 26115 and analyzed according to proposed method.

The results are as under:

Table 2.

Sample No	ΔE
1	2.74
2	2.74
3	2.66
4	2.76
5	2.63
6	2.65
Mean ΔE	2.696

The ΔE values obtained 2.69, its indicates that method can calculate accurately ΔE and discriminate adulteration samples.

Precision- Repeatability

Dye Red Code No. 25880

Control No. 80648XW

This was determined by carrying out Six replicate on the same lot of dye red Code 25880.

The results are reported as under:

Table 3.

Sample No	ΔE
1	0.52
2	0.52
3	0.50
4	0.55
5	0.52
6	0.55
Mean ΔE	0.526
% RSD	0.02 %

Repeatability shows that the method is acceptable and precise, considering the above results the method may be considered Validated.

RESULTS

Table 4.

	L	A	B	ΔE
STANDARD	40.58	44.01	4.96	
TOLERANCE+	NMT 2.0%	NMT 2.0%	NMT 2.0%	
TOLERANCE-	NMT 2.0%	NMT 2.0%	NMT 2.0%	
SAMPLE50	0.25	0.19	0.01	0.45
SAMPLE49	0.25	0.20	0.01	0.46
SAMPLE48	0.25	0.20	0.01	0.46
SAMPLE47	0.25	0.19	0.02	0.46
SAMPLE46	0.25	0.20	0.01	0.46
SAMPLE45	0.25	0.18	0.01	0.44
SAMPLE44	0.25	0.19	0.02	0.46
SAMPLE43	0.25	0.21	0.01	0.47
SAMPLE42	0.25	0.19	0.01	0.45
SAMPLE41	0.25	0.20	0.01	0.46

SAMPLE40	0.25	0.19	0.01	0.45
SAMPLE39	0.25	0.19	0.02	0.46
SAMPLE38	0.23	0.18	-0.01	0.40
SAMPLE37	0.23	0.16	-0.01	0.40
SAMPLE36	0.23	0.17	-0.01	0.39
SAMPLE35	0.23	0.17	0.00	0.40
SAMPLE34	0.23	0.17	-0.01	0.39
SAMPLE33	0.23	0.17	-0.02	0.38
SAMPLE32	0.23	0.17	-0.01	0.39
SAMPLE31	0.23	0.17	-0.01	0.39
SAMPLE30	0.23	0.18	-0.01	0.40
SAMPLE29	0.23	0.17	-0.01	0.39
SAMPLE28	0.23	0.17	0.00	0.40
SAMPLE27	0.23	0.18	-0.01	0.40
SAMPLE26	0.23	0.17	-0.02	0.38
SAMPLE25	0.23	0.18	-0.02	0.39
SAMPLE24	0.23	0.17	-0.01	0.39
SAMPLE23	0.23	0.17	-0.01	0.39
SAMPLE22	0.22	0.18	-0.02	0.38
SAMPLE21	0.23	0.17	-0.01	0.39
SAMPLE20	0.23	0.18	-0.01	0.40
SAMPLE19	0.22	0.18	-0.01	0.39
SAMPLE18	0.23	0.18	-0.01	0.40
SAMPLE17	0.22	0.17	-0.01	0.38
SAMPLE16	0.22	0.17	0.00	0.39
SAMPLE15	0.22	0.18	-0.02	0.38
SAMPLE14	0.23	0.18	-0.02	0.39
SAMPLE13	0.22	0.16	-0.01	0.37
SAMPLE12	0.22	0.17	-0.01	0.38
SAMPLE11	0.22	0.16	0.00	0.38
SAMPLE10	0.22	0.18	-0.01	0.39
SAMPLE9	0.23	0.18	-0.01	0.40
SAMPLE8	0.22	0.18	-0.01	0.39
SAMPLE7	0.22	0.16	0.00	0.38
SAMPLE6	0.21	0.17	-0.01	0.37
SAMPLE5	0.22	0.16	0.00	0.38
SAMPLE4	0.22	0.17	0.00	0.39
SAMPLE3	0.22	0.18	-0.01	0.39
SAMPLE2	0.22	0.18	0.00	0.40
SAMPLE1	0.22	0.17	0.00	0.39

CONCLUSION

The paper describes a new method for determination of coating solutions by reflectance spectroscopy. We analyze

fifty different samples of coating solution and random error found less than 2.0 % this shows reliability of data and high degree of precision under our defined new analytical method for the measurement of coating solution in term of reflectance. The proposed method is rapid, precise and the obtained results are in a good agreement with the declared contents. The accuracy and precision of the method has been confirmed by the statistical parameters.

ACKNOWLEDGEMENTS

Research work is a complex and lengthy process which requires the cooperation of many peoples and institutions .We especially thank of Abbott Laboratories Pakistan Limited, HEJ research institute of Chemistry and all our colleagues, friends, family for their forbearance during the research works.

REFERENCES

- [1] Schweppe, H. 1988. Practical information for the identification of dyes on historic textile materials. Washington, D.C. Smithsonian institution press.
- [2] Young, W. J. 1957. Appendix III: Analysis of textile dyes. Paper of the Peabody Museum of Archaeology and Ethnology 50:53-54.
- [3] Saltzman, M., A. M., Keay, and J., Christensen. 1963. The identification of colorants in ancient textiles. *Dyestuffs* 44:241-51
- [4] Saltzman, M. 1978. The identification of dyes in archaeological and ethnographic textiles. In *Archaeological chemistry II*, ed .G .F. Carter .Advances in chemistry series 171. Washington ,D.C :American Chemical Society. 172-85
- [5] Saltzman, M. 1986. Analysis of dyes in museum textiles; or, you can not tell a dye by its color. In *Textile conservation symposium in honor of Pat Reves*, ed .C. C. McLean and P. Connel. Los Angeles: Conservation centre, Los Angeles County Museum of Art. 32-39.
- [6] Saltzman, M., A. M., Keay, and J., Christensen. 1963. The identification of colorants in ancient textiles. *Dyestuffs* 44:241-51.
- [7] Saltzman, M. 1962. Identifying dyes in textiles .*American Scientist* 80:474-81.
- [8] Abrahams, D. H., and S. M., Edelstein .1964. A new method for the analysis of ancient dyes textiles. *American Dyestuff Reporter* 53:19-25.
- [9] Hofenck-ed Graff, J. H. 1969. Natural dye stuffs: Origin ,chemical constitution identification .In *ICOM Committee for conservation Preprints, 2d plenary Meeting*, Amsterdam . Paris :ICOM. 69/16.
- [10] Martoglio ,P .A., S. P., Bouffard , A. J., Sommer, J. E., Katon ,, and K. A., Jakes. 1990. Unlocking the secrets of the past :The analysis of archaeological textiles and dyes *Analytical Chemistry* 62:1123A-28A.
- [11] Geiss-Mooney ,M. E., and H. L., Needles. 1981. Dye analysis of a group of Late Intermediate period textile from Ica, Peru. In *preservation of paper and textile historic and artistic value II*, ed J . C. Williams. Advance in chemistry series 193. Washington D.C: American Chemical society. 291-300.
- [12] Kharbade ,B. V., and O. P., Agrawal .1985. Identification of natural red dyes in old Indian textiles : Evaluation of thin layer chromatographic systems. *Journal of Chromatography* 347:447-54.
- [13] Schweppe, H. 1986. Identification of dyes in historic textile materials. In *historic textile and paper materials: Conservation and characterization*, ed .H. L .Needles and S. H. Zeronian .Advance in chemistry series 212 .Washington, D.C: American Chemical Society. 153-74.
- [14] Wouters, J., and N. Rosario-Chirinos. 1992. Dye analysis of pre-Columbian Peruvian textiles with high performance liquid chromatography and diode-array detection .*Journal of the American Institute for Conservation* 31:237-55.
- [15] Wouters, J., 1985 .High performance liquid chromatography of anthraquinones: Analysis of plants and insect extract and dyed textiles. *Studies in Conservation* 30:119-28.
- [16] Fischer ,Ch. H., M., Bischof, and J. G., Rabe .1990. Identification of natural and early synthetic textile dyes with HPLC and UV/VIS- spectroscopy by diode array detection *Journal of Liquid Chromatography* 13:319-31.
- [17] Marttogoio, P. A., Bouffard S. P., Sommer A. J., Katon ,J. E. and Jakes K. A., 1990. Unlocking the secrets of the past :The analysis of archeological textiles and dyes. *Analytical Chemistry* 62:1123A-28A.
- [18] Geoffroy ,H. and Town ,A. D .1997 .Dyes drive from amino thiophen ,part 4 : synthesis of some nitro substitutes thiophen based azo dispersed dyes 33(4). 319-336.
- [19] Modi, B. J and Patel, R. G. 1998. Synthesis of Noval azo dispersed dyes containing benzimidazole ring system for dyeing nylon and polyester fiber ,*Test dye printer*, 31(8), 12-16.
- [20] Synthesis of 5-chloro-2-hydroxy-1-naphthyl azo volume 55 ,abstract number 97100a year 1947-1956 .
- [21] Lake Red SK synthesis of ,volume 53 abstract number 11842g year 1957-1961.
- [22] Formation from dyes and metal hydroxole. 61;11369f 1962-1966 .
- [23] Lakes (pigment); for cosmetic ,food & pharmaceutical chem., legal and phys. aspect of 67;R 120107v 1967-1971
- [24] Titanium dye oxide, for tablets coating ,73;59274c 1967-1971.
- [25] Lake 170(55069-14-2) 82;141654w 1972-1976.
- [26] Emulsifiers ,for emulsion polmn. of butadiene with itaconic acid styrene ,particle size in relation to 87;p 137088a 1977-1982.
- [27] Dyes & Pigments book 319 ,Absorbent & luminescence of silver particle coated with 104;195825u .1982 -1986 & 1987-1991.
- [28] The pigment are indexed at Dyes and pigment ,see dyes & pigment , chemical Abstract 1992-1996 .
- [29] Lake Red (69344-93-0) 131-SC 6 Vol. 131 Book 2246 . 1999 year.
- [30] Lake sediment of quinone profile ,212299y chem. abstract vol. 132, 2000 .
- [31] Typical weighing factors are given by ASTM 258-7-1-1951 as reported in the journal of the optical society of America Vol 41, 1951 pages 431-439.